33.1: Sponges are basal animals that lack tissues

Branch point 1: First level-1 branch: Porifera. The second level-1 branch splits from branch point 2 into the first level-2 branch: Cnidaria. The second level-2 branch splits from branch point 3 into the first and second level-3 branches. The first level-3 branch splits from branch point 4 into the first level-4 branch: Lophotrochozoa and the second level-4 branch: Ecdysozoa. The second level-3 branch is Deuterostomia.

Animals in the phylum Porifera are known informally as sponges. Among the simplest of animals, sponges are sedentary and were mistaken for plants by the ancient Greeks. Most species are marine, and they range in size from a few millimeters to a few meters. Sponges are **filter feeders**: They filter out food particles suspended in the surrounding water as they draw it through their body, which in some species resembles a sac perforated with pores. Water is drawn through the pores into a central cavity, the **spongocoel**, and then flows out of the sponge through a larger opening called the **osculum(Figure 33.3)**. More complex sponges have folded body walls, and many contain branched water canals and several oscula.

Figure 33.3

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Anatomy of a sponge.
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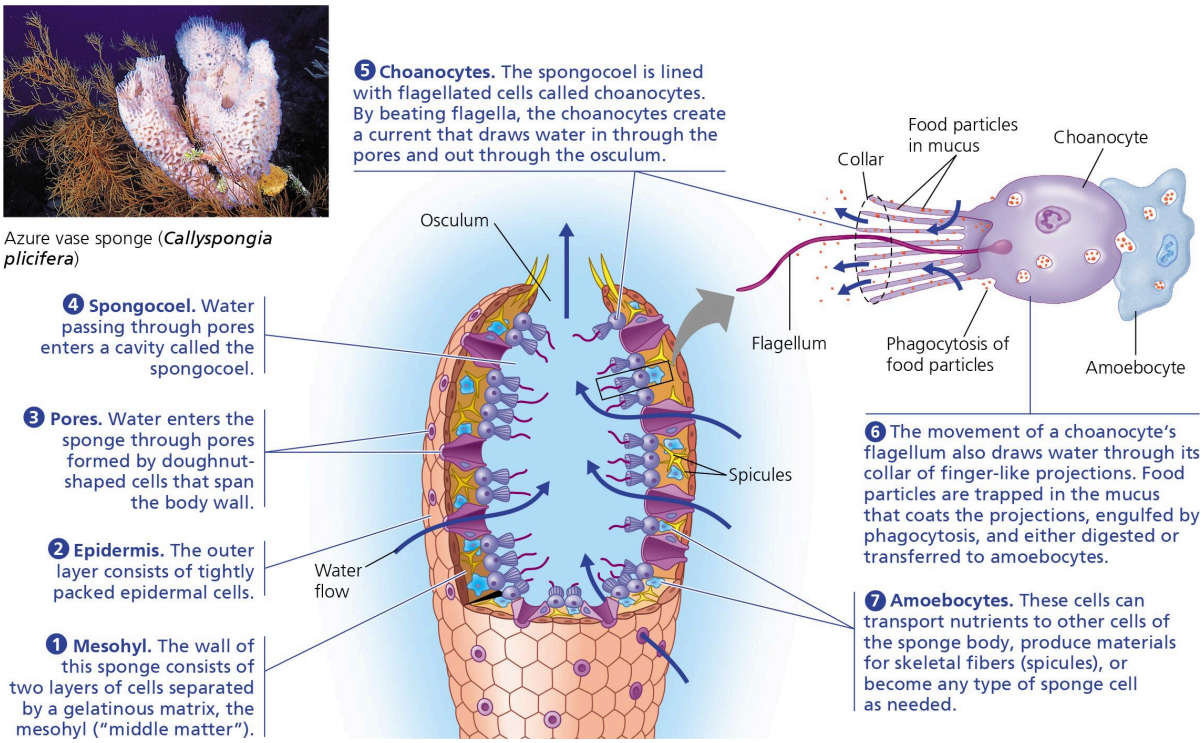

In the main diagram, portions of the front and back wall are cut away to show the sponge's internal structure.

Seven features of sponge anatomy include:

1.Mesophyll. The wall of this sponge consists of two layers of cells separated by a gelatinous matrix, the mesophyll (middle matter).

2.Epidermis. The outer layer consists of tightly packed epidermal cells.

3.Pores. Water enters the sponge through pores formed by doughnut-shaped cells that span the body wall.

4.Spongocoel. Water passing through pores enters a cavity called the spongocoel.

5.Choanocytes. The spongocoel is lined with flagellated cells called choanocytes. By beating flagella, the choanocytes create a current that draws water in through the pores and out through the osculum.

6.The movement of a choanocyte‘s flagellum also draws water through its collar of finger-like projections. Food particles are trapped in the mucus that coats the projections, engulfed by phagocytosis, and either digested or transferred to amoebocytes.

7.Amoebocytes. These cells can transport nutrients to other cells of the sponge body, produce materials for skeletal fibers (spicules), or become any type of sponge cell as needed.

Sponges represent a lineage that diverged from other animals early in the history of the group; thus, they are said to be *basal animals*. Unlike nearly all other animals, sponges lack tissues, groups of similar cells that act as a functional unit as in muscle tissue and nervous tissue. However, the sponge body does contain several different cell types. For example, lining the interior of the spongocoel are flagellated **choanocytes**, or collar cells (named for the finger-like projections that form a "collar" around the flagellum). These cells engulf bacteria and other food particles by phagocytosis. The similarity between choanocytes and the cells of choanoflagellates supports molecular evidence indicating that animals and choanoflagellates are sister groups (see Figure 32.3).

The body of a sponge consists of two layers of cells separated by a gelatinous region called the **mesohyl**. Because both cell layers are in contact with water, processes such as gas exchange and waste removal can occur by diffusion across the membranes of these cells. Other tasks are performed by cells called **amoebocytes**, named for their use of pseudopodia. These cells move through the mesohyl and have many functions. For example, they take up food from the surrounding water and from choanocytes, digest it, and carry nutrients to other cells. Amoebocytes also manufacture tough skeletal fibers within the mesohyl. In some sponges, these fibers are sharp spicules made from calcium carbonate or silica. Other sponges produce more flexible fibers composed of a protein called spongin; you may have seen these pliant skeletons being sold as brown bath sponges. Finally, and perhaps most importantly, amoebocytes are *totipotent* (capable of becoming other types of sponge cells). This gives the sponge body remarkable flexibility, enabling it to adjust its shape in response to changes in its physical environment (such as the direction of water currents).

Most sponges are **hermaphrodites**, meaning that each individual functions as both male and female in sexual reproduction by producing sperm *and* eggs. Almost all sponges exhibit sequential hermaphroditism: They function first as one sex and then as the other. Cross-fertilization can result when sperm released into the water current by an individual functioning as a male is drawn into a neighboring individual that is functioning as a female. The resulting zygotes develop into flagellated, swimming larvae that disperse from the parent sponge. After settling on a suitable substrate, a larva develops into a sessile adult.

Sponges produce a variety of antibiotics and other defensive compounds, which hold promise for fighting human diseases. For example, a compound called cribrostatin isolated from marine sponges can kill both cancer cells and penicillin-resistant strains of the bacterium *Streptococcus*. Other sponge-derived compounds are also being tested as possible anticancer agents.